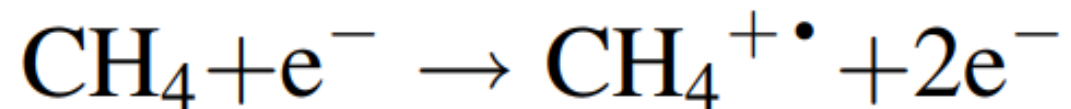
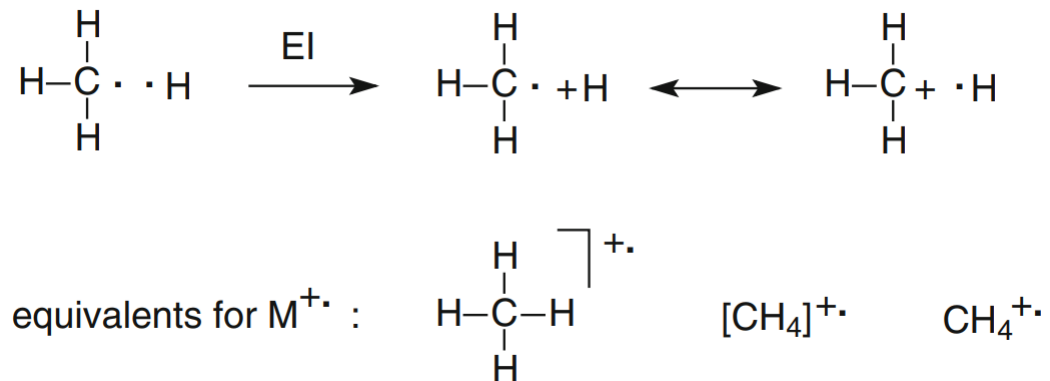
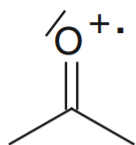


Interpretation of EI Mass Spectra

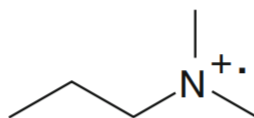


The tremendous changes in electronic structure and bonding that arise with the loss of an electron are by far better depicted by a formula representation including the σ -bonding electrons of the C–H bond.

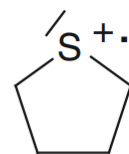




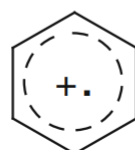
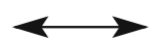
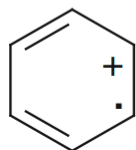
acetone



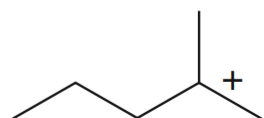
N,N-dimethyl-propylamine



tetrahydrothiophene



Benzene



2-methyl-1-pentene

几个重要的知识点

- 质谱反应基本上属于**单分子反应**；
- 双分子反应和成簇分子反应少（主要原因是离子源中样品的蒸汽压很低）；
- 由于带有电荷，分子离子具有较宽的**内能**；
- 处于激发态的离子产生各种与能量相关的**分解反应**；
- 每一个分解反应主要产生**一个离子**和一个**中性碎片**（中性丢失，包括不带电荷的自由基）；
- 离子丰度的主要影响因素是：（1）产生**离子的稳定性**，和（2）产生**中性离去碎片的倾向性**。
- 离子化能量的需求的强度大致和UV中电子跃迁的顺序一致： $\sigma > \pi > n$

Even-Electron Rule

- It serves as a reliable guideline rather than a strict rule for the fragmentation of ions, and thus for the interpretation of mass spectra.

Even-electron rule

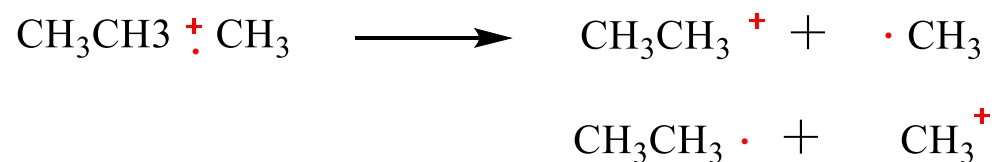
- Odd-electron ions (OE, such as molecular ions and fragment ions formed by rearrangements) may eliminate either a radical or an even-electron neutral species, but even-electron ions (EE, such as protonated molecules or fragment ions formed by a single bond cleavage) will not usually lose a radical to form an odd-electron cation. In other words, the successive loss of radicals is forbidden。



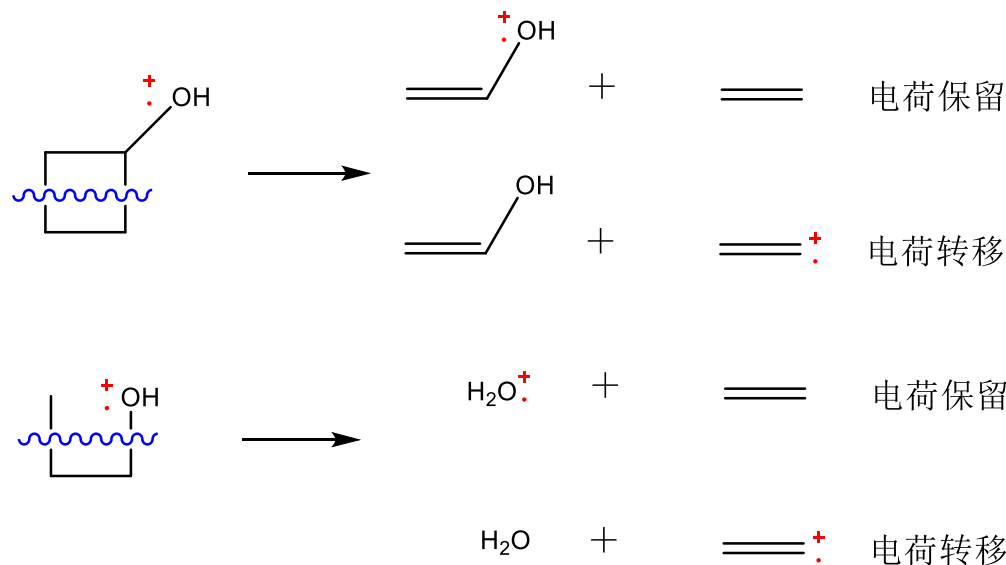
反应的类型

奇电子分子离子:

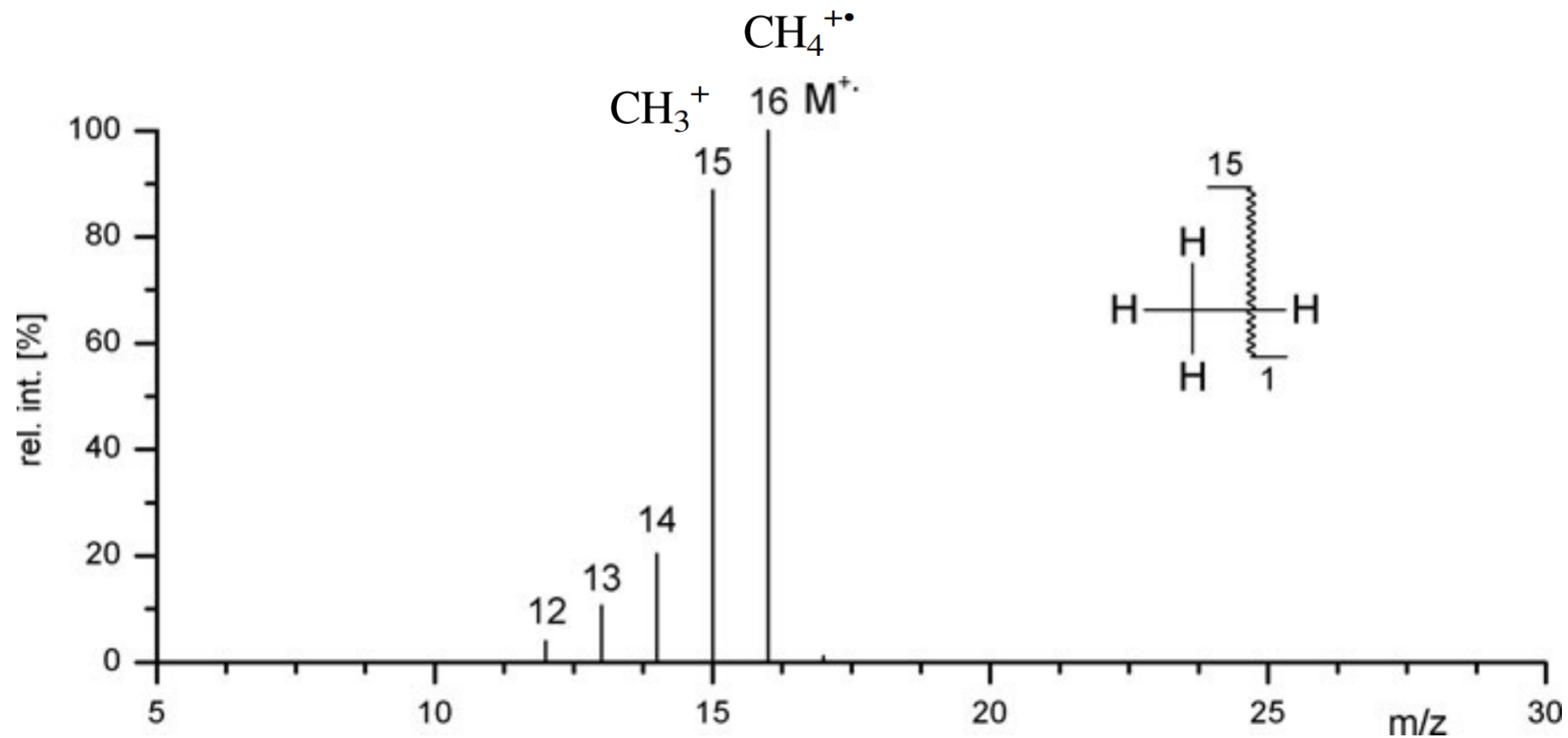
- 如果只断裂一个键, 则产生一个偶电子离子和一个中性碎片;



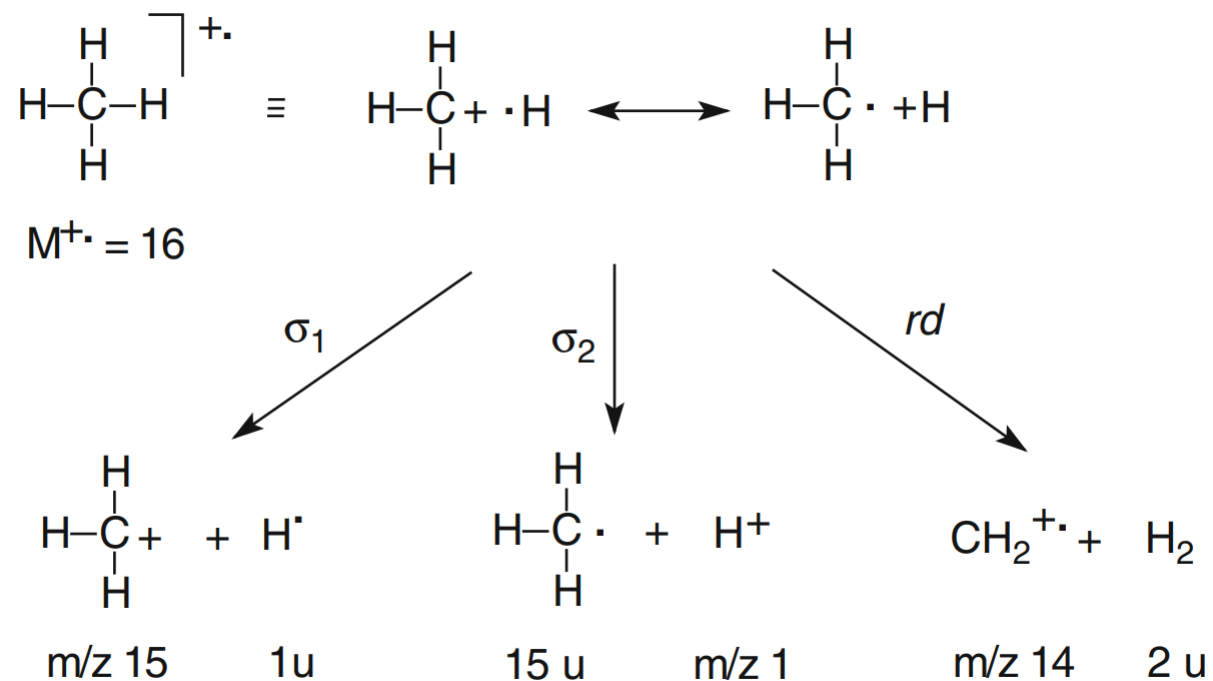
- 如果断两个键 (涉及开环和重排反应), 则产生奇电子碎片;



σ -Bond Cleavage in Small Nonfunctionalized Molecules



The charge and radical state are then attached to the brackets

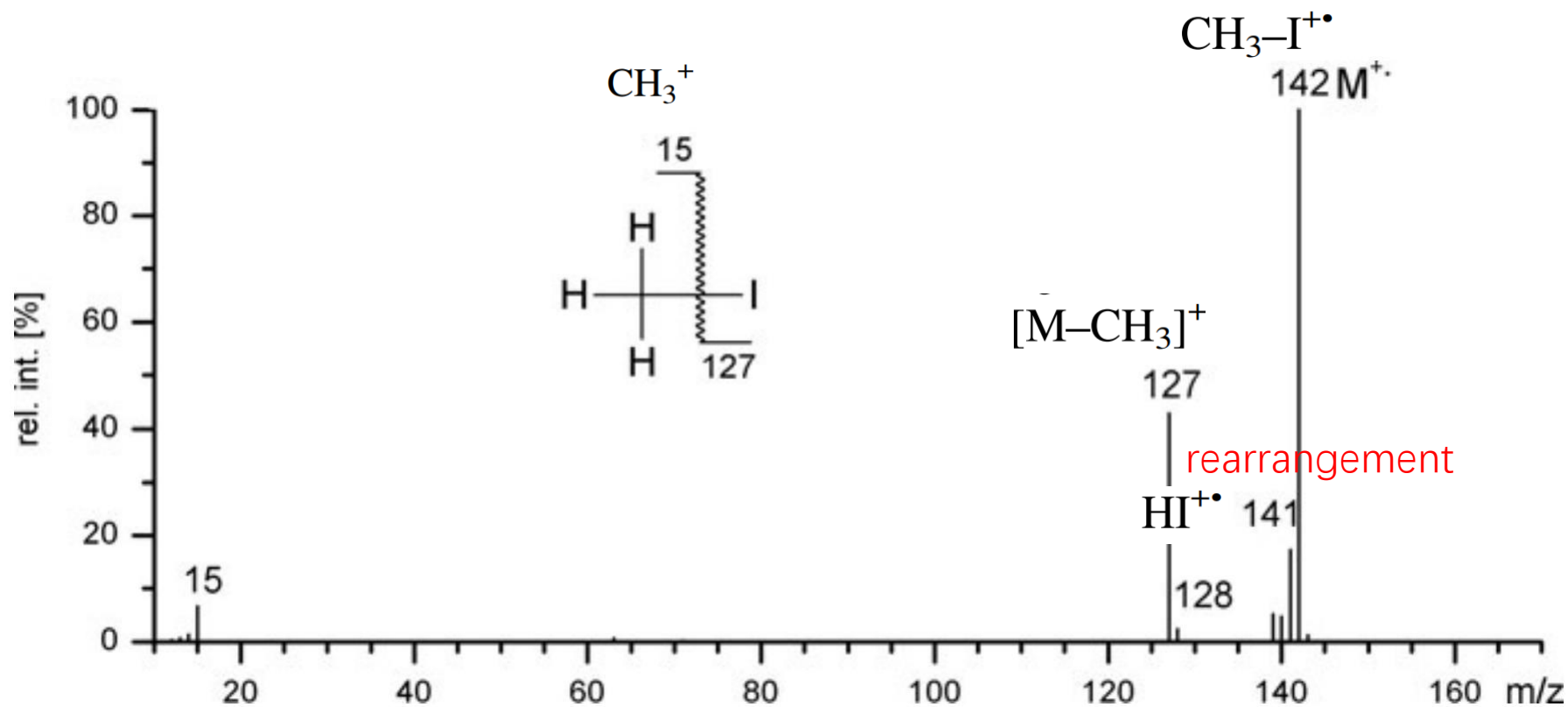


氮规则：

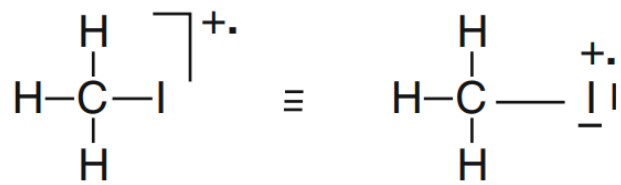
在只含C、H、O、N的化合物中，不含或含偶数个氮原子的分子的质量数为偶数，含有奇数个氮原子的分子的质量数为奇数。这是因为在由C、H、O、N、P、卤素等元素组成的有机分子中，只有氮原子的化合价为奇数而质量数为偶数

氮原子数量	示例	M+•的m/z
0	甲烷, CH ₄	16
0	丙酮, C ₃ H ₆ O	58
0	氯仿, CHCl ₃	118
0	[60]富勒烯, C ₆₀	720
1	氨, NH ₃	17
1	乙腈, C ₂ H ₃ N	41
1	吡啶, C ₅ H ₅ N	79
1	N-乙基-N-甲基丙胺, C ₆ H ₁₅ N	101
2	脲, CH ₄ N ₂ O	60
2	吡嗪, C ₄ H ₄ N ₂	80
3	三唑, C ₂ H ₃ N ₃	69
3	六甲基磷酰三胺, HMPTA, C ₆ H ₁₈ N ₃ OP	179

σ -Bond Cleavage in Small Functionalized Molecules



Note that the peak at m/z 128 should not be interpreted as an isotopic peak of m/z 127, because iodine is monoisotopic

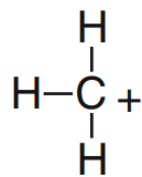


$M^{+\cdot} = 142$

σ_1

σ_2

rd

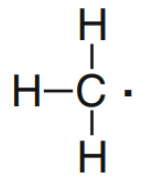


+



m/z 15

127 u

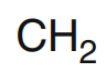


+



15 u

m/z 127



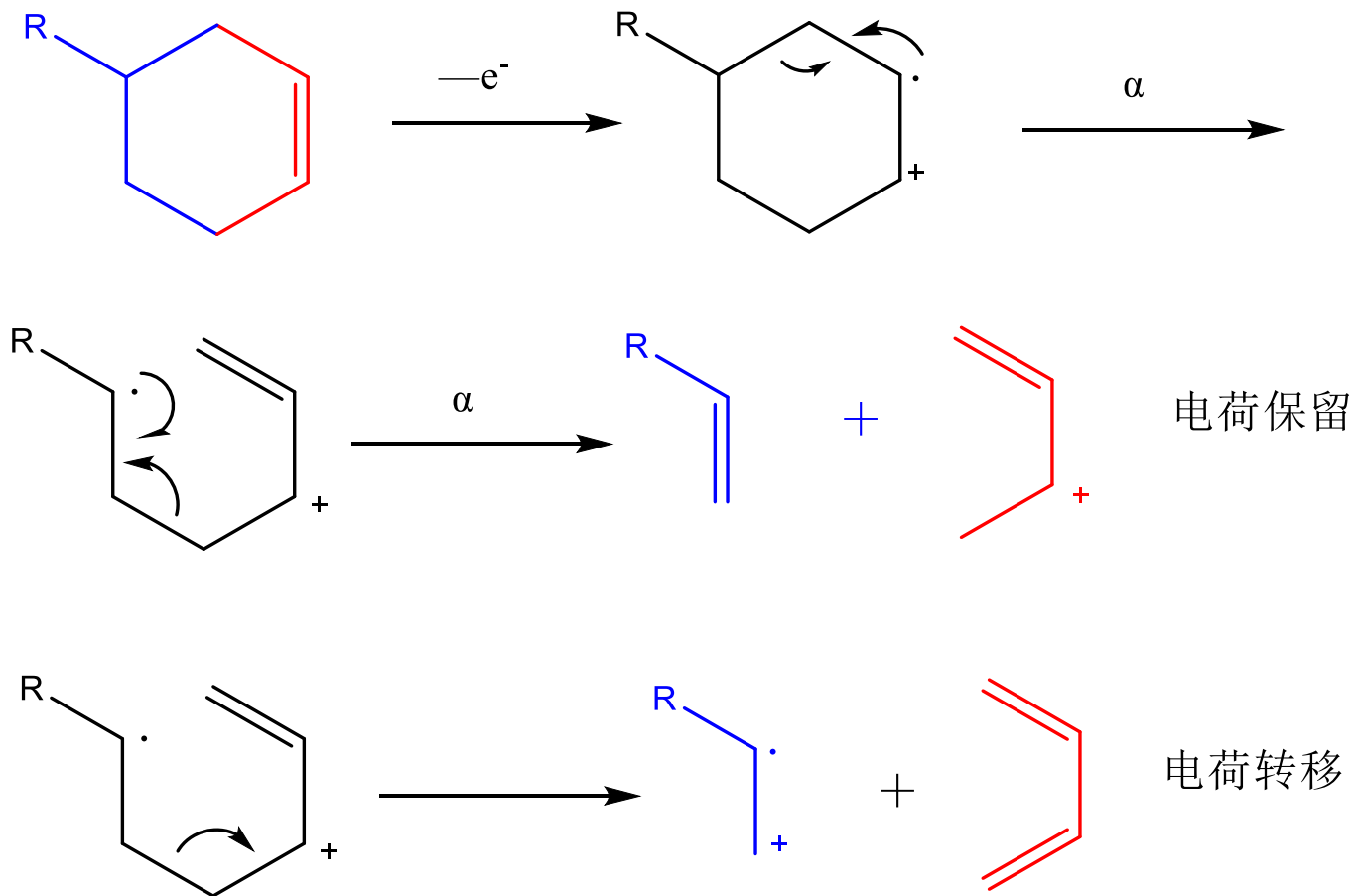
+



14 u

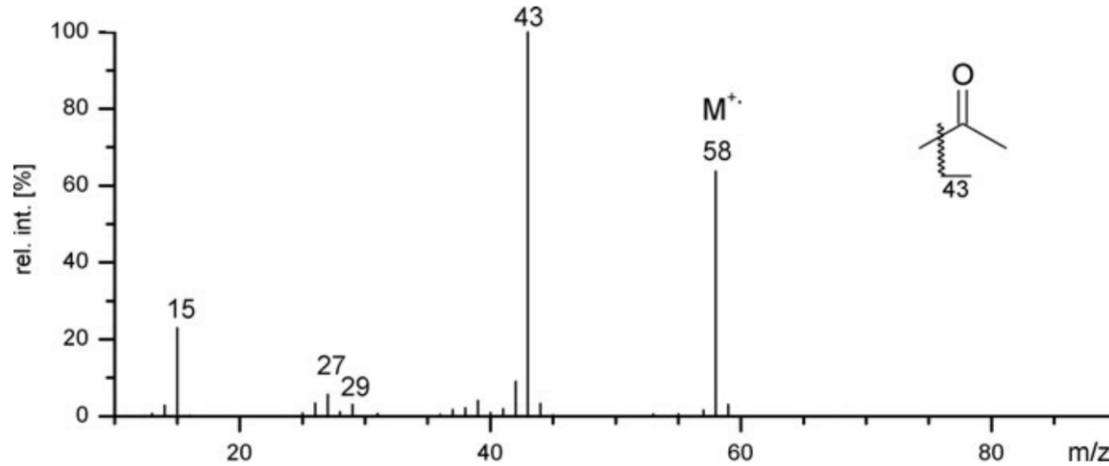
m/z 128

(电荷中心) α 裂解, 以反-DA反应为例

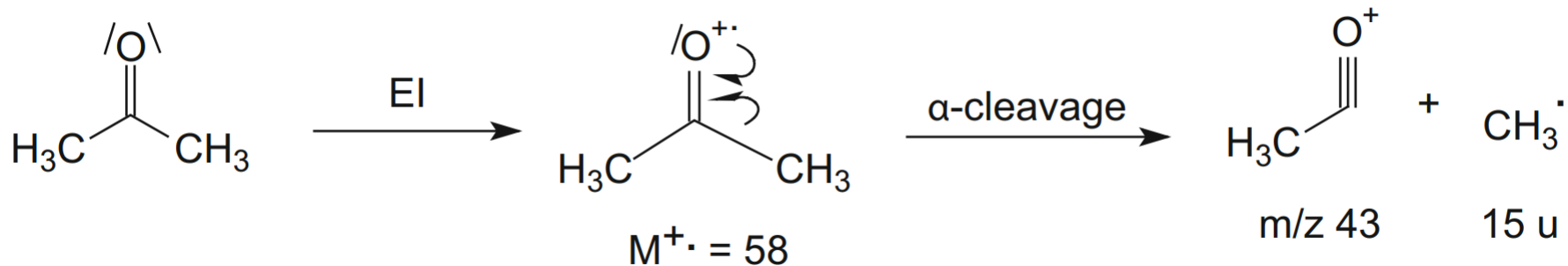


Alpha-Cleavage

在奇电子离子中，与其相邻原子的外侧键断裂，属于该原子的一个电子转移，并与游离基中心的未成对电子形成新键，构成较稳定的碎片离子。

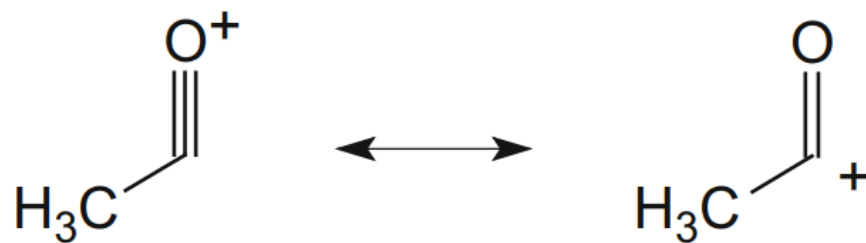


The excited molecular ion may then cleave off a methyl radical by simply shifting one electron from the CO-CH₃ bond to the radical site at the oxygen atom.

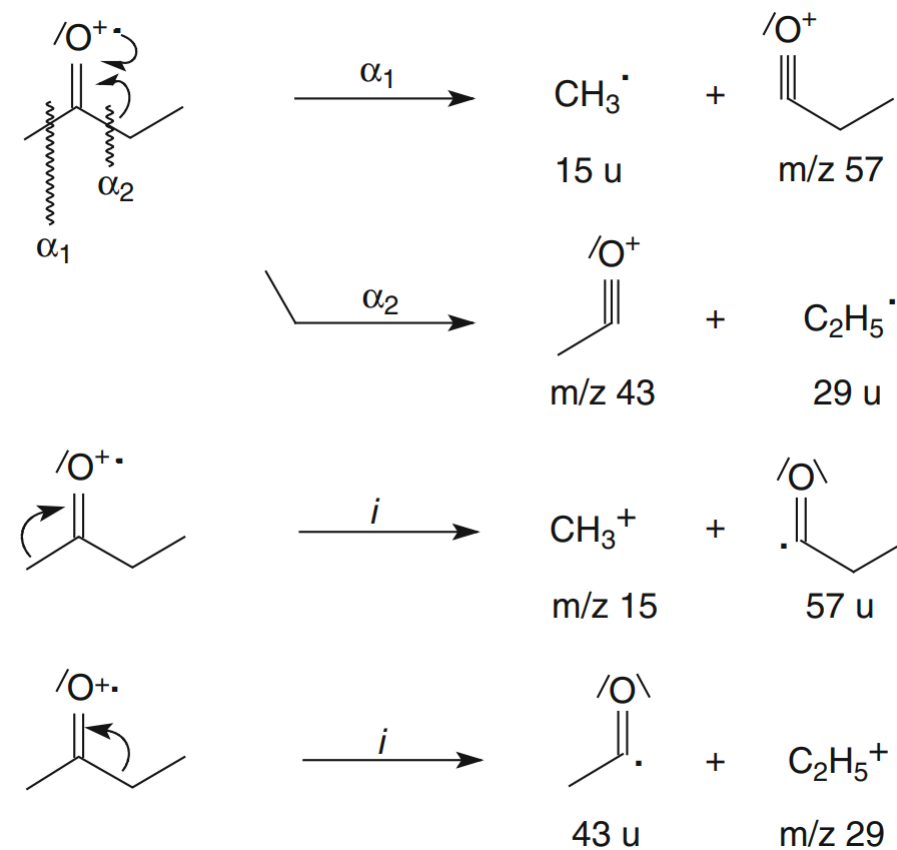
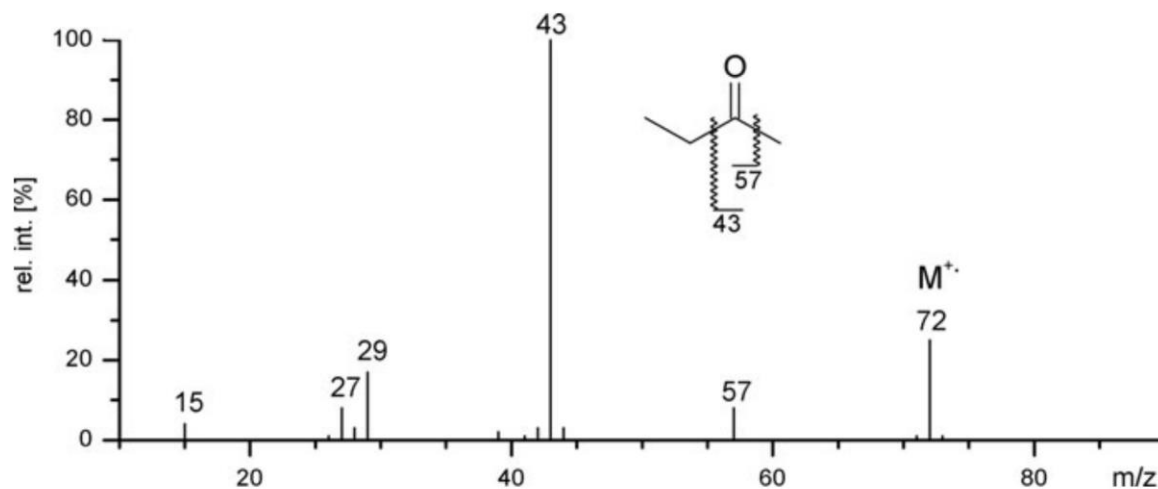


The term α -cleavage for this widespread radical-site initiated process with charge retention can be misleading, because the bond cleaved is not directly attached to the radical site, but to the next neighboring atom.

The charge in acylium ions can be substantially resonance-stabilized.

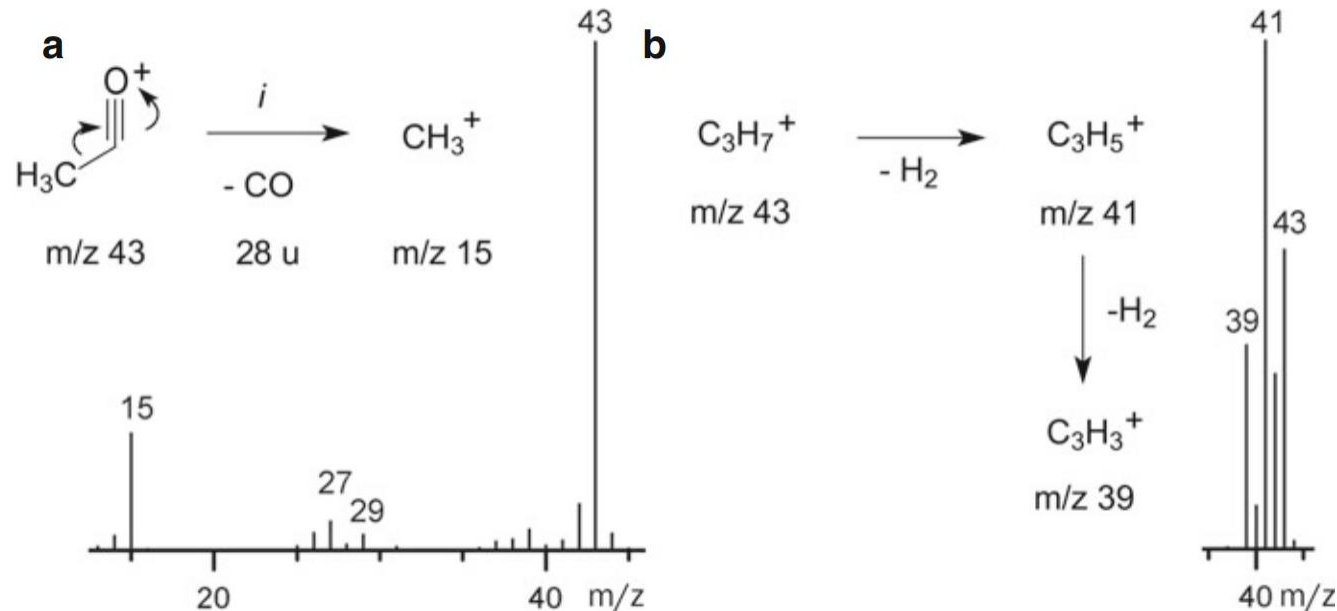


α -Cleavage of Nonsymmetrical Aliphatic Ketones



Acylium Ions and Carbenium Ions

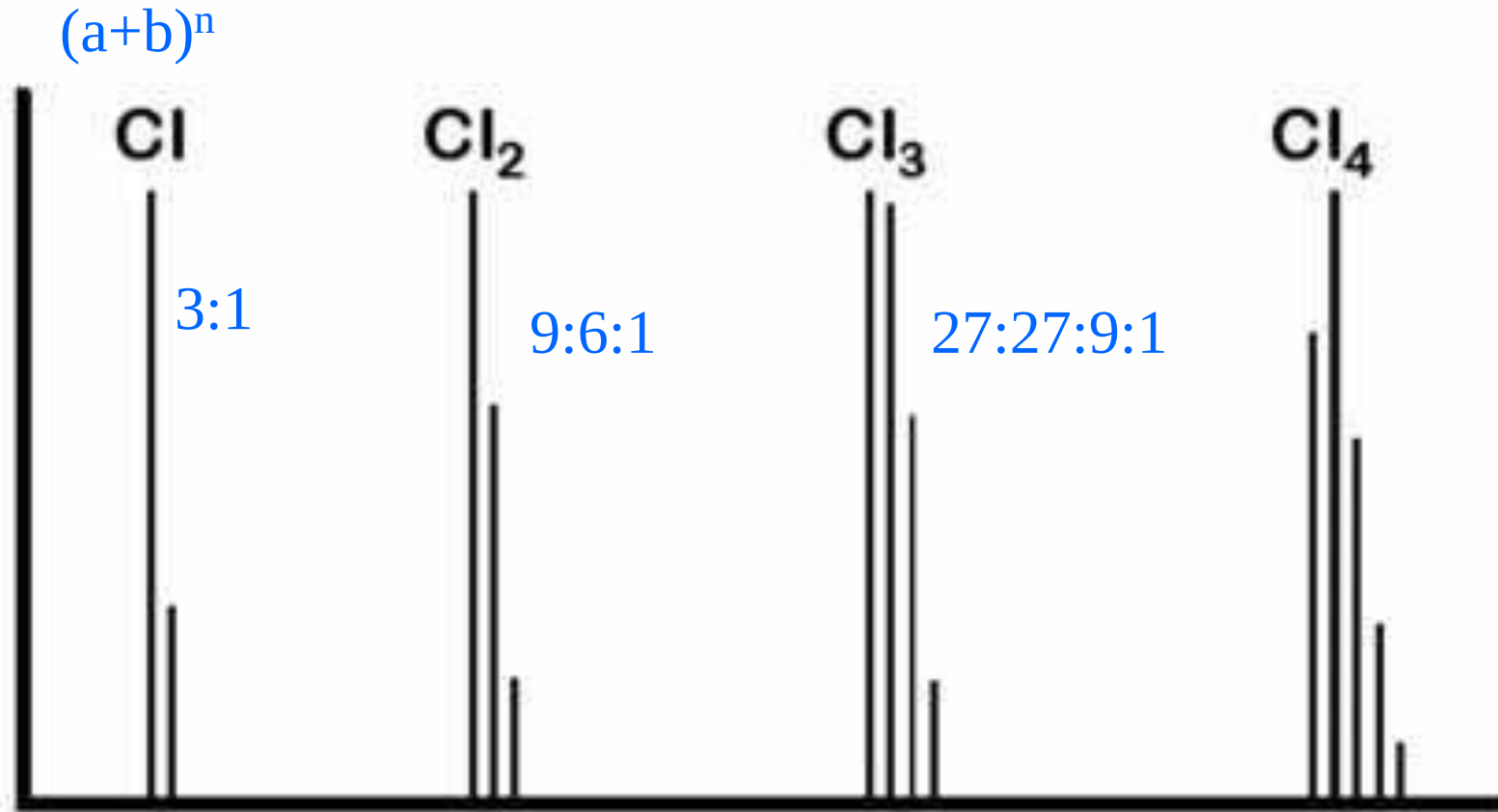
- The acylium ions and the saturated carbenium ions are isobaric ions because they share the same nominal mass. However, their exact masses are different, i.e., CH_3CO^+ , m/z 43.0178, and C_3H_7^+ , m/z 43.0542.



Looking for Hal⁺

- Halogenated compounds typically reveal the positive halogen ion and a less intense peak due to the **hydrogen halogenide**. In accordance with the incrementing electronegativities of the halogens their intensities follow the order **F⁺ < Cl⁺ < Br⁺ < I⁺**.
- 对于碳原子来说，¹²C的天然丰度为98.9%，¹³C的天然丰度为1.07%，从M+1峰与M峰强度的比值，可估算出分子中含碳的数目，碳原子的数目上限 = $I(M+1)/I(M) \div 1.1\%$ ，式中I(M+1)和I(M)分别表示M+1峰和M峰的强度。
- 因此，含有一个碳原子的化合物如甲烷，M=17与M=16的峰强度之比应该为0.011.其他元素也有同样的规律，含有一个氯原子的化合物中， $I(M+2)/I(M)=32.5\%$ ，而在含有一个溴原子的化合物中， $I(M+2)/I(M)=1$ 。
- 对于含有C、Br、Cl等同位素天然丰度较高的元素的化合物，其同位素离子峰相对强度可由 $(a+b)^n$ 展开式计算，式中a、b分别为该元素轻、重同位素的相对丰度，n为分子中该元素个数。
- 在含有两个氯原子的化合物中，a=3，b=1，n=2，故 $(a+b)^2=9+6+1$ ，则其分子离子峰与相应同位素离子峰强度之比为9：6：1。

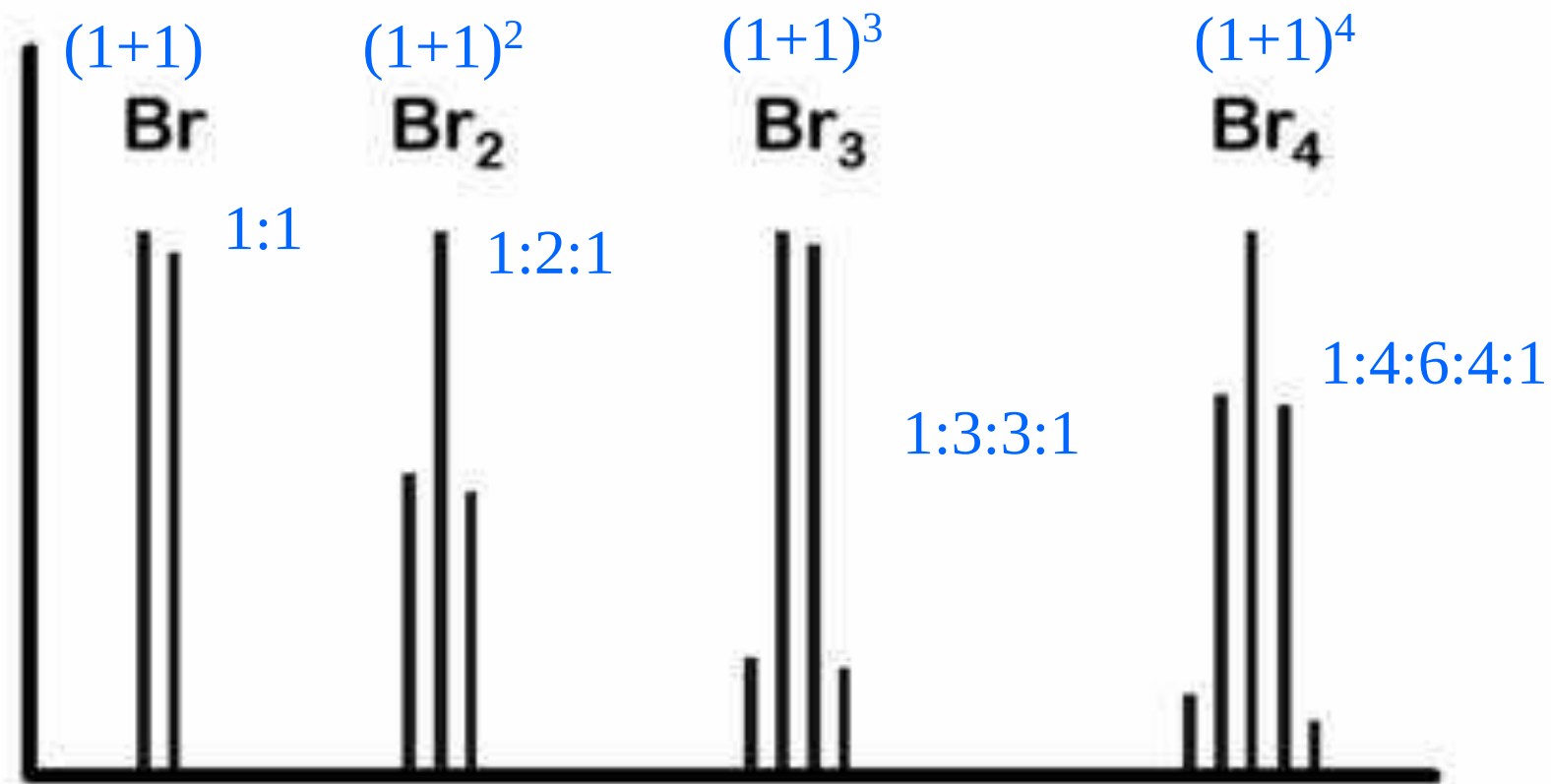
Chlorine



By the time you have 4 chlorine, the A+2 line is significantly larger than the A line.

Bromine

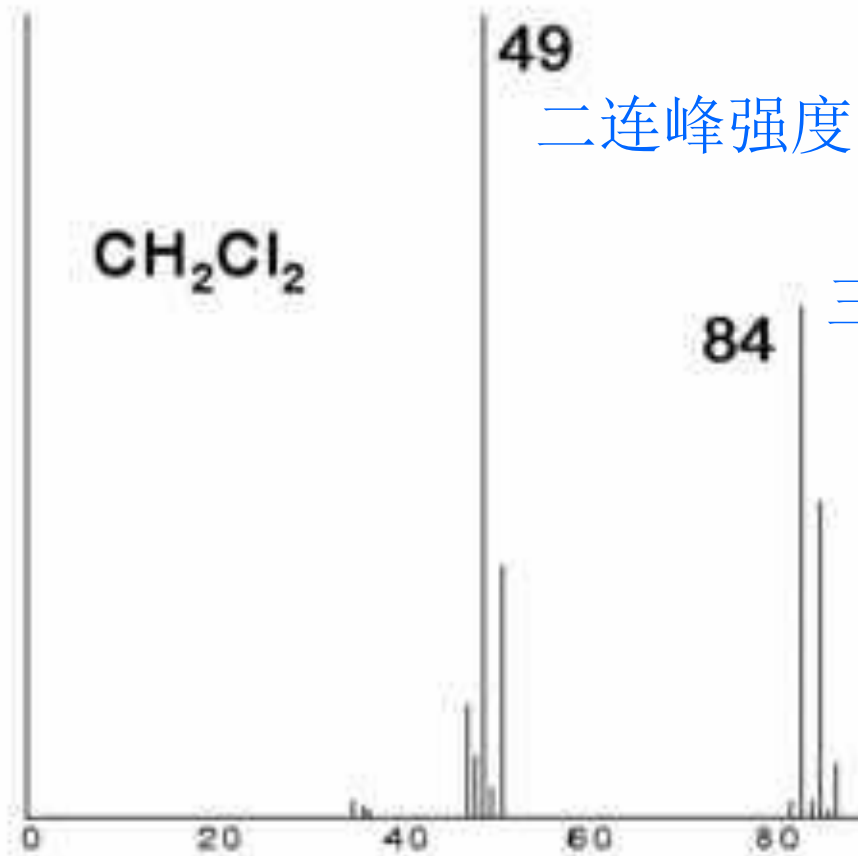
$^{81}\text{Br}/^{79}\text{Br}=97.9:100 \sim 1:1$



Bromine, which is almost a 1:1 mixture of ^{79}Br and ^{81}Br also has a distinctive pattern.

Another example

Dichloromethane - a common GC/MS solvent



二连峰强度比为 3:1

三连峰强度比为 9: 6: 1

Note the Cl_2
pattern at 84
and the Cl
pattern at 49

